

of P , we get that $\mathbf{0}$ and $\mathbf{1} - \mathbf{e}_i$ have distance $d-1$ in $G(F_{(i)})$, so by induction on d we get

$$F_{(i)} = \text{conv}\{\mathbf{x} \in \{0, 1\}^d : x_i = 0\}.$$

Collecting all the vertices that we now know have to be in P , we get $P = \text{conv}(\{0, 1\}^d) = I_d$ and $\delta(G(P)) = d$. $\square$

The bound $\delta(G(P)) \leq \dim(P)$ can also be proved (with the same kind of argument) in the monotone version, where we ask for the shortest path “to the top” with respect to a given linear function. If we restrict to the strictly monotone version of Conjecture 3.9, then the characterization of the equality case also remains valid (with the same proof).

3.4 Kalai's Simple Way to Tell a Simple Polytope from Its Graph

In this section we consider simple polytopes and their graphs. Our treatment is based on a striking (and strikingly simple) paper by Gil Kalai. To be honest — the situation is even worse: the following is copied quite directly from his paper “**A simple way to tell a simple polytope from its graph**” [299].

Let P be a simple d -dimensional polytope and let $G(P)$ be the graph of P . Thus, $G(P)$ is an abstract graph defined on the set of vertices $\text{vert}(P)$ of P . Two vertices $\mathbf{v}$ and $\mathbf{u}$ in $\text{vert}(P)$ are adjacent in $G(P)$ if $[\mathbf{v}, \mathbf{u}]$ is a one-dimensional face of P .

Perles [435] conjectured the following result.

Theorem 3.12. (Blind & Mani [108])

If P is a simple polytope, then the graph $G(P)$ determines the entire combinatorial structure of P .

In other words, if two simple polytopes have isomorphic graphs, then their face lattices are isomorphic as well.

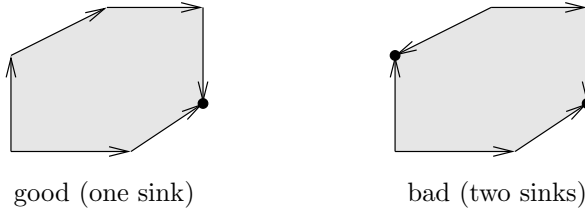
Proof. Here is Kalai's [299] simple proof of this result.

We consider the set of *all* acyclic orientations (i.e., edge orientations with no oriented cycles) of $G(P)$. We will not distinguish between an orientation O of $G(P)$ and the partial order induced by O on $\text{vert}(P)$, which is defined by $\mathbf{v} \leq_O \mathbf{u}$ whenever there is an O -directed path from $\mathbf{v}$ to $\mathbf{u}$.

Note that if O is an acyclic orientation of $G(P)$, then the induced subgraph obtained by restriction of $G(P)$ to any nonempty subset A of $\text{vert}(P)$ has a sink (an element with out-degree zero) with respect to O .

An acyclic orientation O of $G(P)$ is called *good* if for every nonempty face F of P , the graph $G(F)$ has exactly one sink. Otherwise, O is *bad*.

The existence of good acyclic orientations of $G(P)$ follows from Theorem 3.7: if $\mathbf{c}\mathbf{x}$ is in general position for P , then it is also for all faces of P . Our first goal is to distinguish intrinsically between good and bad orientations of $G(P)$.



Let O be an acyclic orientation of $G(P)$. Let h_k^O be the number of vertices of $G(P)$ with in-degree k in O . Define

$$f^O := h_0^O + 2h_1^O + 4h_2^O + \cdots + 2^k h_k^O + \cdots + 2^d h_d^O.$$

If $\mathbf{x}$ is a vertex of $G(P)$ of in-degree k with respect to O , then $\mathbf{x}$ is a sink in 2^k faces of P . (Since P is simple, every i edges incident to $\mathbf{x}$ determine an i -face F of P which includes them.) Let f denote the number of nonempty faces of P . Since each face has at least one sink, we see that

- I.) $f^O \geq f$, and
- II.) O is good if and only if $f^O = f$.

To distinguish between good and bad orientations from the knowledge of $G(P)$ only, compute f^O for every acyclic orientation O . The good acyclic orientations of $G(P)$ are those having the minimal value of f^O .

Now we will show how to identify the faces of P . The graphs of simple k -polytopes are k -regular: they have exactly k edges incident to every vertex, by Proposition 2.16. With this, the criterion is very simple: an induced connected k -regular subgraph H of $G(P)$ is the graph of some k -face of P if and only if its vertices are initial with respect to some good acyclic orientation O of $G(P)$. Indeed, if F is a face of P , it is well known that $\text{vert}(F)$ is an *initial set* for some good acyclic orientation: a set of vertices such that no directed edge leads into the set. For this just consider a linear function with respect to which the vertices of F lie below all other vertices, which can be obtained by choosing a linear function $\mathbf{c}\mathbf{x}$ that defines F , and perturbing it according to Lemma 3.4.

On the other hand, let H be a connected k -regular subgraph of $G(P)$ and let O be a good acyclic orientation with respect to which $\text{vert}(H)$ is an initial set. Let $\mathbf{x}$ be a sink of H with respect to O . There are k edges containing $\mathbf{x}$ in H , all oriented toward $\mathbf{x}$. Therefore $\mathbf{x}$ is a sink in the k -face F that contains these k edges. Since the orientation O is good, $\mathbf{x}$ is the unique sink of F , and therefore all vertices of F are $\leq \mathbf{x}$, with respect to O . But $\text{vert}(H)$ includes the set of all vertices that are $\leq \mathbf{x}$ with

respect to O . (Remember: $\text{vert}(H)$ is an initial set with respect to O .) Thus, $\text{vert}(F) \subseteq \text{vert}(H)$. Since both H and $G(F)$ are k -regular and connected, $\text{vert}(F) = \text{vert}(H)$ and $G(F) = H$. This completes the proof. $\square$

Remarks 3.13.

0. You could ask: do these parameters h_i^O actually *mean* anything? They do: we will come back to this when we study shellability, in Lecture 8.
1. We do not have a practical way to distinguish between good and bad orientations. The algorithm suggested by the preceding proof is exponential in $|\text{vert}(P)|$, but it can be made to “work” in practice: see Achatz & Kleinschmidt [1]. Is there a really *efficient* way to compute, for example, the number of facets of P from $G(P)$?
2. General polytopes cannot be reconstructed from their graphs — this can be seen, for example, from the existence of neighborly (simplicial) polytopes. However, Joswig [293] has an extension of Kalai’s result and proof to non-simple polytopes.

Perles [435, 437] proved that simplicial d -polytopes are determined by their $\lfloor d/2 \rfloor$ -skeleta; see Kalai [307]. General d -polytopes are determined by their $(d-2)$ -skeleta, and this is best possible even for quasi-simplicial polytopes (all of whose facets are simplicial polytopes); see Grünbaum [252, Ch. 12].

3.5 Balinski's Theorem: The Graph is d -Connected

A very fundamental fact about the graphs of d -polytopes is that they are d -connected, a theorem due to Balinski [37].

Here we use the definition that a simple graph $G(P)$ is *d -connected* if the removal of any $d-1$ or fewer vertices (and all the edges they are incident with) leaves a connected graph.

The theorem is certainly plausible, since it is easy to see (using Lemma 3.6) that every vertex of $G(P)$ has degree at least d . We have adapted the following simple proof from Grünbaum [252]. Two different proofs appear in Brøndsted & Maxwell [134] and in Barnette [50]. Two extensions, which answer the questions of how many components the graph may have if you remove k vertices, or if you remove a k -face, appear in Klee [322] and in Perles & Prabhu [439]. A stronger, “directed” version of Balinski’s theorem is hidden in Holt & Klee [282].

Theorem 3.14 (Balinski’s theorem). [37]

The graph $G(P)$ is d -connected for every d -polytope P .

Proof. Let $P = \text{conv}(V) \subseteq \mathbb{R}^d$, where the vertex set V of P and of the graph $G(P)$ has at least $d+1$ elements. We delete a subset of $d-1$ of them, $S = \{v_1, \dots, v_{d-1}\} \subseteq V$; then we have to show that the graph $G(P) \setminus S$ induced on the remaining vertices is connected.